

Smart devices call for a smart design

Experts offer six design tips to keep in mind when developing embedded electronics for everyday products

Good performance within energy budgets. Embedded electronics must be power efficient. The importance of this parameter may be ascertained by the number of power-specific benchmark tests that have evolved over the past years. "Even there are instances where such devices are also responsible for generating power to meet their needs—a case in this point is the pacemaker that generates power from body heat. Instances of devices that are self-sufficient relative to power are on the rise," explains Vivek Tyagi, country sales director, Freescale Semiconductor. There is no single power-reduction technique that is able to meet all the system requirements for energy minimisation. The trick is to effectively combine architectural, platform and circuit techniques, system and application software, process technology and design methodology and tools to intelligently develop semiconductor designs that deliver the highest possible performance levels within a restricted energy budget.

Safety first, style next. Engineers also need to be very careful about safety issues. "As a result of electronics performing minute activities for humans, the solutions tend to be overstressed and might end up causing accidents or critical damage. In order to determine how safe a system needs to be, you have to look at the risk for a given event that will lead to the hazard. In addition, you have to take into account the probability of exposure to this event and the potential severity. These steps will lead to the safety requirements of the system. Once the safety requirements are defined, the designer then has to determine how to reduce any resulting risk of all potential hazards below the acceptable risk level. The use of redundant systems for increasing the safeness of a system is standard procedure, but the design and implementation of the redundancy often requires some creativity. Electronics systems are not an exception," explains Tyagi.

"The electronics embedded in everyday objects need to be ergonomic, and electromagnetic radiations need to be kept in control. Keeping the system error proof and durable, along with an appropriate operating temperature, needs to be considered when making electronics for daily use," says Satish Mohanram, technical marketing manager, National Instruments.

Dr Arun Mulpur, manager-Industry Marketing, MathWorks Inc., points out that the designers must also ensure the safety and privacy of personal data, preferences and behaviours that their creations are going to process, enable and monitor.

Pre-emptive steps are critical to make a design fool-proof and safe for general use. For example, in the case of the self-stirring mug cited earlier by T. Anand, principal consultant, Knewron, he says that the insulation design (and its quality) between the hot contents of the cup and the batteries, electronics hardware, etc can make or break the

product. If not handled well, it could lead to accidents and injuries.

Nihal Kashinath, founder, IoTBLR explains that in addition to the obvious safety factors of the device itself, some of the risks come from external sources and are often not thought of during product design. Network security (especially when such products are connected to the Web and become vulnerable to hacker attacks), fire hazards (if they come in contact with flammable material in the environment, e.g., a leaking gas cylinder in the kitchen), materials used (toxic plastics, lead components, etc) and even human behaviour—the way humans would react to the use of an electronic device, are some aspects to look into. "Case in point: women security bracelets. While the intention is good, I am yet to come across a single design that actually provides safety against an attacker's reaction when the woman triggers an SOS using the bracelet. It could possibly infuriate the attacker (or worse, a group of attackers who may be intoxicated and violent) and provoke them to do something even more drastic before any help can arrive," alerts Kashinath.

Keep it simple. T. Anand adds, "For industrial designs, the user profile is always known (such as technical staff, workers, engineers, etc). However, for mass market consumers, the designer may not know the end-user profile. This means the product must be simple to use. It does not mean it can't be sophisticated and complex, rather it means that the user experience should be simple (no matter what is available inside)."

Close your eyes and see the user. The Electronics of Things is a science that is all about people and their use of technology. So, always keep the user in mind. "While designing embedded electronics for everyday objects, engineers must remember to consider aesthetic, ergonomic, functional and usability aspects. Besides, careful study must be done to understand cognitive and emotional aspects so that user experience is positive," says Ugarkar.

Look ahead and see the Rol. Ugarkar also warns that, "The addition of electronics must not complicate the manufacturing process or alter the reliability of the object or increase the price beyond a reasonable limit. Rol remains the biggest challenge in this space." Tyagi also points out that costing is a very important element in the design of these systems and is a key component of the decision-making process throughout the product life cycle.

Make the product self-aware. "How good a design is in terms of self-awareness would become important in the coming days when electronics becomes more pervasive," says Anand. For instance, if the battery life is over, the gadget should be intelligent enough to indicate so and refuse to function. If mistakes are made by the user while operating, it should resurrect itself and avoid any mishaps.

director, SAP Labs India. "Notably, we have smart meters, smart cards and smart accessories. Smart meters not only track the consumption but also send that data over wireless networks to the data centre on a continuous basis. Imagine an entire city being monitored by smart meters and then these meters sending information to a central station for billing purposes. At the end of the month, the bill is automatically generated and sent to the consumer via email and balance

due is deducted from the bank account without requiring any human intervention. That is the age we are walking into! The amount of silicon that is getting installed into the modern-day cars is also mind-boggling. It not only calculates the optimal travel route but also monitors the health of the car from engine status to wheel pressure. The intelligence built into the chips can predict the service date and sets up an appointment at a nearby service centre for the same. Again, without any hu-

man intervention! The latest in silicon technology is what we call wearable devices. These modern accessories not only look fashionable but also have some serious brains behind them. They can monitor the heart rate, blood pressure, calories burnt, etc and pass on that information to smartphones for the purpose of analysis. The day is not far when the smart accessories you wear can predict your next ailment and automatically notify your physician for a check-up!" he explains.